

Evaluation of Quality of Obturation Using Two Different Rotary Files and Hand Files in Primary Teeth: A Randomized Controlled Trial

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INTRODUCTION

The evolution in dentistry has led to several developments in recent years. This had led to improvisation not just in the material aspect but also in the instrumentation and technique aspects leading to better work quality particularly in fields like pulp therapy.¹

Pulpectomy involves the removal of infected tissue in symptomatic deciduous teeth which is considered a taxing procedure in the domain of pediatric dentistry.² The success of a pulpectomy relies on the obturation of the root canal system in the same way as the apt method of instrumentation and chemo-mechanical preparation.³

It is said that the length of the treatment procedure strongly influences the child's behavior and the conventional manual instrumentation method used for cleaning as well as shaping proves to be long and cumbersome.^{4,5} There have been several studies documented on the usage of hand files in primary root canals for instrumentation.⁶ Despite their wide usage, these hand files have few undeniable limitations which include lengthy canal preparation time as well as unequal preparation of root canals.⁴

At the start of the year 2000, NiTi files were introduced into the field of Pediatric Endodontics by Barr et al.⁶ Rotary instrumentation using NiTi files thus has gained popularity in recent years to overcome the disadvantages of manual instrumentation. Rotary files are preferred for their ability to reduce the instrumentation time, conserve the sinuous root canal path of deciduous teeth, and also provide conical-shaped canals enhancing the quality of obturation.^{4,7,8}

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Very few studies have been done comparing the different rotary file systems in primary teeth.^{1,5,7-14} Systematic clinical trials need to be conducted with rotary files in primary teeth to validate the results and to arrive at definitive conclusions. There are no *in vivo* studies done comparing the quality of obturation and time taken for instrumentation of Kedo-SG blue and ProTaper Gold Rotary Files in deciduous teeth. Therefore, this particular study was conducted to compare and evaluate instrumentation time and quality of obturation of ProTaper gold and Kedo-SG rotary files with hand files in primary molars. The hypothesis is the presence of a significant difference in the instrumentation time and obturation

quality using rotary instrumentation when compared with manual instrumentation in deciduous molars.

Glass ionomer cement (Shofu, Shofu Inc., Japan) was provided as the entrance filling and SS crowns were used for restoration either on the same day or on the next day of the appointment.

MATERIALS AND METHODS

The randomized controlled trial was conducted in the Department of Pediatric and Preventive Dentistry in dental college from the month of July to November of 2018. Approval for trial design was obtained from the Institutional Review Board and got informed consent from both the parents of the children who took part in the study.

Forty-five children in total aging 4–8 years who required pulpectomy in any of the deciduous mandibular molars were allocated to one of the three groups: Group I: manual hand K files, group II: ProTaper Gold rotary files, and group III: Kedo-SG blue rotary files. An individual not involved in this study prepared the computer-generated randomization sequence. We arrived at a sample size of 45 utilizing G power analysis from a previous *in vivo* study with 95% power.⁸ Children were selected for this study if they met the following criteria: (a) vital or non-vital lower deciduous molars without sinus tract, (b) without external or internal pathologic root resorption in teeth, (c) existence of adequate coronal tooth structure to support stainless steel crown. Those children exhibiting a lack of cooperative ability, children having underlying systemic diseases, and children requiring special healthcare needs were excluded from the study.

A single operator carried out all the procedures. A full-mouth examination with intraoral periapical radiographs of all the teeth indicated for pulpectomy were taken before the start of the clinical procedure. After diagnosis was confirmed, local anesthesia 2% lignocaine with 1:200,000 adrenaline (LOX*2% adrenaline, Neon Laboratories Limited, India) was administered. The tooth was isolated with the help of a rubber dam (GDC Marketing, India).

A round carbide bur in a high-speed handpiece was used to remove superficial caries and also derooft the pulp chamber. Coronal pulp amputation was carried out using a spoon excavator and a no. 10 size K-file was utilized to determine the canals' patency. Working length determination was by Ingle's radiographic method using the preoperative radiograph. The canal was prepared using:

Group I: Hand K-files (Mani, Tochigi, Japan) from #15 to #30 employing a quarter pull turn method.

Group II: Hand K file #15, #20 followed by ProTaper Gold file-S2 (Dentsply India Pvt. Ltd., Delhi, India) up till the working length with help of X-Smart endodontic motor.

Group III: Hand k file #15 followed by Kedo-SG Blue-D1, E1 was used till the working length with help of X-Smart motor (Reeganz dental care Pvt. Ltd. India).

The canals' instrumentation time was recorded in seconds by an assistant using a stopwatch. Saline irrigation was done and then dried using sterile paper points. Calcium hydroxide and iodoform paste were used for obturation by gently pushing with help of cotton pellets (Metapex, Meta Biomed Co., Ltd., Korea). A post-obturation radiograph was taken to evaluate the obturation quality. It was then analyzed by another pediatric dentist who was blinded to the type of instrumentation employed.

Based on this, the obturation quality was graded as optimal fill, overfill, and underfill (modification by Coll JA and Sadrian R):¹⁵

Under-filling: ZOE was >2 mm short of the apex in all the canals.

Optimal filling: ZOE ends at the radiographic apex or up to 2 mm short of the apex in one or more of the root canals.

Over-filling: ZOE outside the root in any canal.

LIMITATION

A possible limitation in the current study was that only mandibular molars were taken into consideration for the study.

CONCLUSION

The obturation quality is more acceptable with rotary files than the manual instrumentation method. The minimization of instrumentation time in the pulpectomy procedure was also noticed with rotary file use in deciduous molars. On that account, Kedo-SG blue or ProTaper Gold rotary file systems can be used as an optional technique for root canal preparation in deciduous teeth.

REFERENCES

1. Crespo S, Cortes O, Garcia C, et al. Comparison between rotary and manual instrumentation in primary teeth. *J Clin Pediatr Dent* 2008;32(4):295–298. DOI: 10.17796/jcpd.32.4.I57I36355u606576.
2. Bowen JL, Mathu-Muju KR, Nash DA, et al. Pediatric and general dentists' attitudes toward pulp therapy for primary teeth. *Pediatr Dent* 2012;34(3):210–215.
3. Endo MS, Ferraz CCR, Zaia AA, et al. Quantitative and qualitative analysis of microorganisms in root-filled teeth with persistent infection: monitoring of the endodontic retreatment. *Eur J Dent* 2013;7(3):302–309. DOI: 10.4103/1305-7456.115414.
4. Silva LAB, Leonardo MR, Nelson-Filho P, et al. Comparison of rotary and manual instrumentation techniques on cleaning capacity and instrumentation time in deciduous molars. *J Dent Child Chic Ill* 2004;71(1):45–47.
5. Manual and rotary instrumentation techniques for root canal preparation in primary molars | Rosa | *Dentistry 3000* [Internet]. [cited 2019 Feb 21]. Available from: <http://dentistry3000.pitt.edu/ojs/index.php/dentistry3000/article/view/19>.

6. Barr ES, Kleier DJ, Barr NV. Use of nickel-titanium rotary files for root canal preparation in primary teeth. *Pediatr Dent* 2000;22(1):77–78.
7. Pinheiro SL, Araujo G, Bincelli I, et al. Evaluation of cleaning capacity and instrumentation time of manual, hybrid and rotary instrumentation techniques in primary molars. *Int Endod J* 2012;45(4):379–385. DOI: 10.1111/j.1365-2591.2011.01987.x.
8. Govindaraju L, Jeevanandan G, Subramanian E. Clinical evaluation of quality of obturation and instrumentation time using two modified rotary file systems with manual instrumentation in primary teeth. *J Clin Diagn Res JCDR* 2017;11(9):ZC55–ZC58. DOI: 10.7860/JCDR/2017/30069.10602.
9. Azar MR, Safi L, Nikaein A. Comparison of the cleaning capacity of Mtwo and Pro Taper rotary systems and manual instruments in primary teeth. *Dent Res J* 2012;9(2):146–151. DOI: 10.4103/1735-3327.95227.
10. Katge F, Chimata VK, Poojari M, et al. Comparison of cleaning efficacy and instrumentation time between rotary and manual instrumentation techniques in primary teeth: an in vitro study. *Int J Clin Pediatr Dent* 2016;9(2):124–127. DOI: 10.5005/jp-journals-10005-1347.
11. Ochoa-Romero T, Mendez-Gonzalez V, Flores-Reyes H, et al. Comparison between rotary and manual techniques on duration of instrumentation and obturation times in primary teeth. *J Clin Pediatr Dent* 2011;35(4):359–363. DOI: 10.17796/jcpd.35.4.8k013k21t39245n8.
12. Jeevanandan G. Kedo-S paediatric rotary files for root canal preparation in primary teeth – case report. *J Clin Diagn Res JCDR* 2017;11(3):ZR03–ZR05. DOI: 10.7860/JCDR/2017/25856.9508.
13. Kummer TR, Calvo MC, Cordeiro MMR, et al. Ex vivo study of manual and rotary instrumentation techniques in human primary teeth. *Oral Surg Oral Med Oral Pathol Oral Radiol Endod* 2008;105(4):e84–e92. DOI: 10.1016/j.tripleo.2007.12.008.
14. Govindaraju L, Jeevanandan G, Subramanian EMG. Comparison of quality of obturation and instrumentation time using hand files and two rotary file systems in primary molars: a single-blinded randomized controlled trial. *Eur J Dent* 2017;11(3):376–379. DOI: 10.4103/ejd.ejd_345_16.
15. Coll JA, Sadrian R. Predicting pulpectomy success and its relationship to exfoliation and succedaneous dentition. *Pediatr Dent* 1996;18(1):57–63.
16. Fuks AB, Papagiannoulis L. Pulpotomy in primary teeth: review of the literature according to standardized criteria. *Eur Arch Paediatr Dent Off J Eur Acad Paediatr Dent* 2006;7(2):64–71. DOI: 10.1007/BF03320817discussion 72.
17. Berutti E, Chiandussi G, Gaviglio I, et al. Comparative analysis of torsional and bending stresses in two mathematical models of nickel-titanium rotary instruments: ProTaper versus ProFile. *J Endod* 2003;29(1):15–19. DOI: 10.1097/00004770-200301000-00005.
18. Berutti E, Negro AR, Lendini M, et al. Influence of manual preflaring and torque on the failure rate of ProTaper rotary instruments. *J Endod* 2004;30(4):228–230. DOI: 10.1097/00004770-200404000-00011.